PARISHRAM 2.0

CLASS 12TH 2026

Botany-Sexual Reproduction in Flowering Plants

Biology

LECTURE-3

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TOPICS TO BE COVERED

1. Ovule
2. Pollen Grain
3. Megasporogenesis
4.

Pollen grain
(Terminologies)

Pollen allergy :- Some plant species

- ✓ → Amaranthus
- ✓ → Parthenium / carrot grass / congress grass
- ✓ → Chenopodium

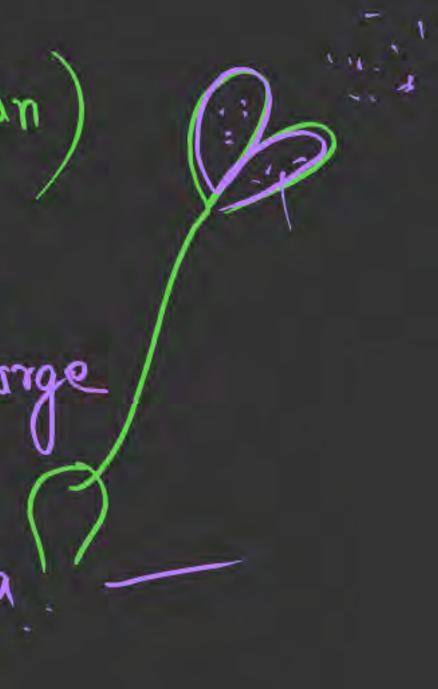
Pollen grains can trigger allergic responses eg sneezing, redness coughing, asthma, bronchial affliction, inflamation of respiratory tract

Pollen nutrient :- Some species pollen grains are rich in nutrient

∴ used for making syrups, tablets & given to race horses | athletes to inc their performance.

Pollen viability → (lifespan)

It is the duration of pollen grain after discharge from anther till landing on stigma.



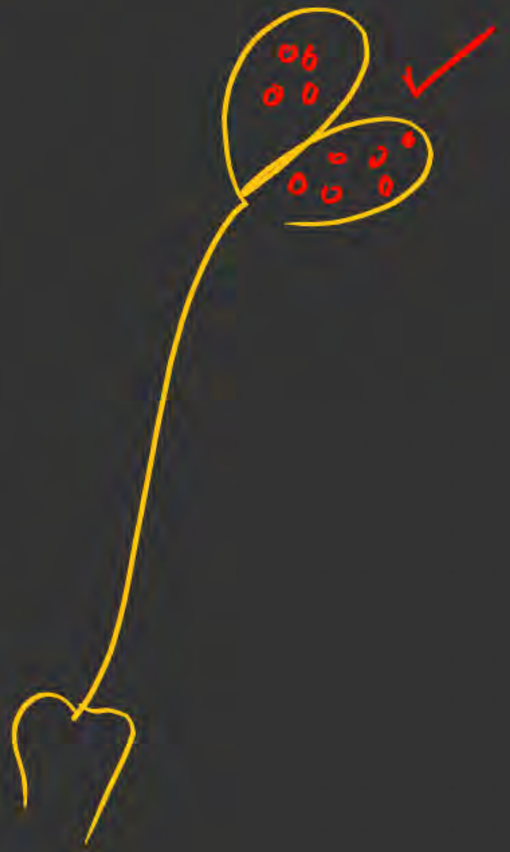
eg - Poaceae
eg Rice / Wheat / Bajra etc
↳ 30 min

↳ Leguminaceae (Pea)
Solanaceae (potato)
Rosaceae
} → Several months

Pollen viability depends upon temp, humidity etc

Pollen bank :- It is the place where pollens are stored for future use in liq N_2 at $-196^{\circ}C$ [Cryopreservation] ✓

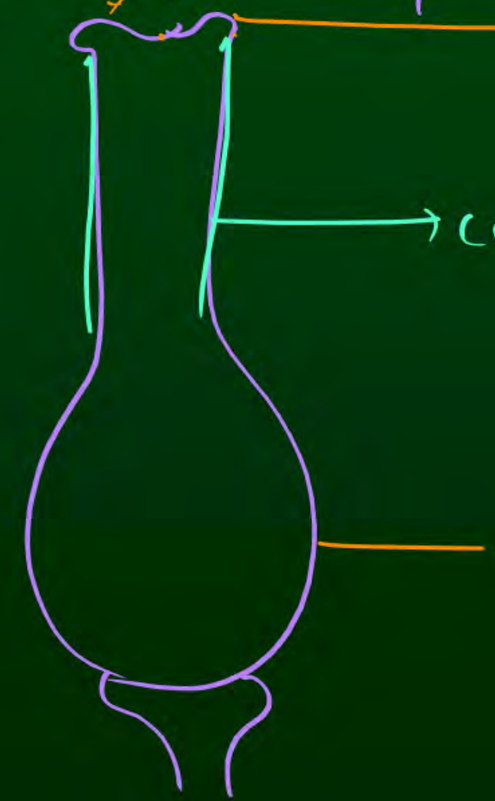
low temp



Carpel;Megasporangium

Female sex organ → Gynoecium [single :- carpel]

Typically carpel :- 3 parts



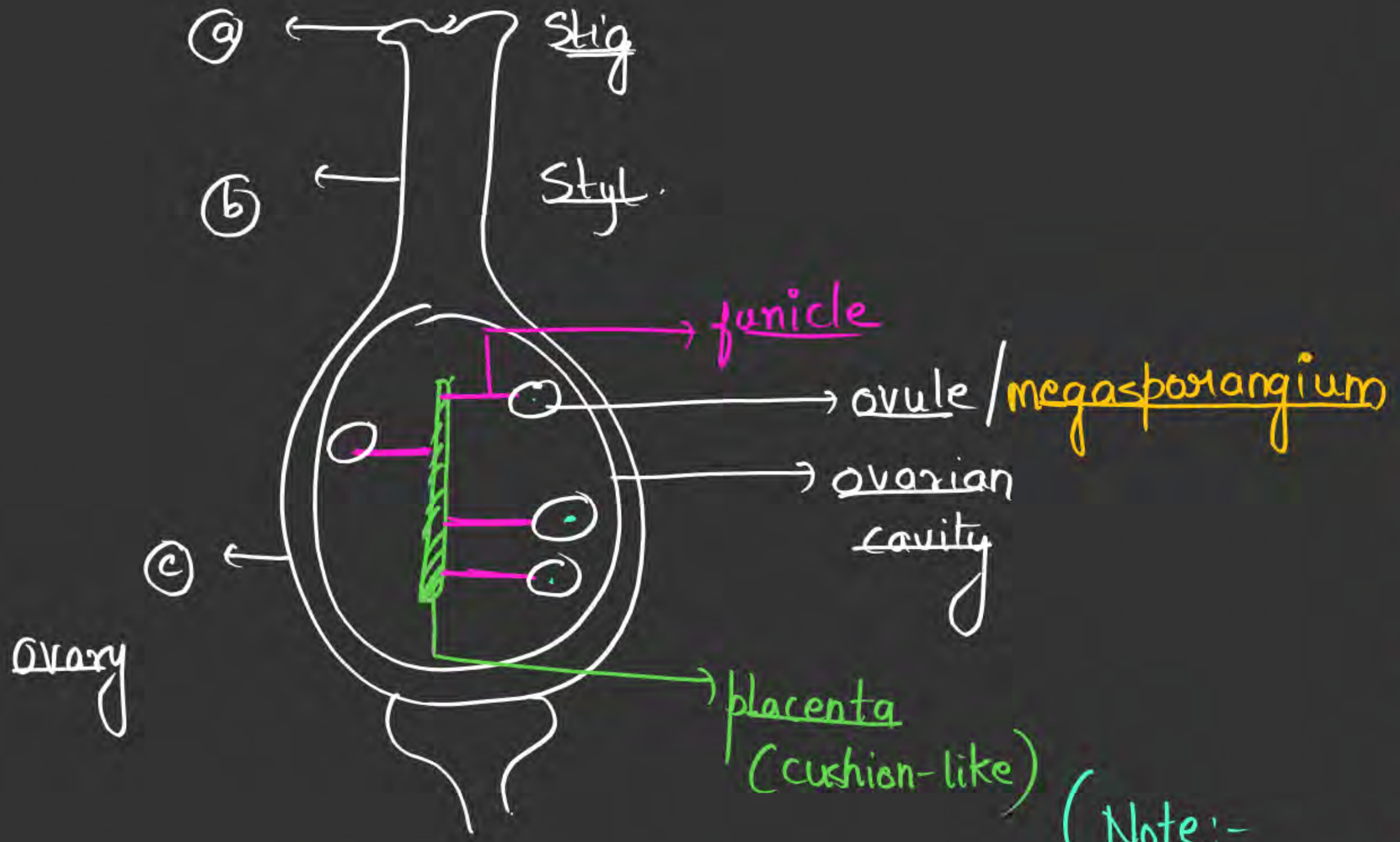
→ landing place of pollengrain

Stigma

→ connecting tube = style

Flower

- one carpel (monocarpellary)
- many carpel (multicarpellary)
  - free (Apocarpous) eg lotus, rose
  - fused (Syncarpous) eg Mustard, pea



Ovary
↓
cavity
↓
placenta
↓
funicle
↓
ovule

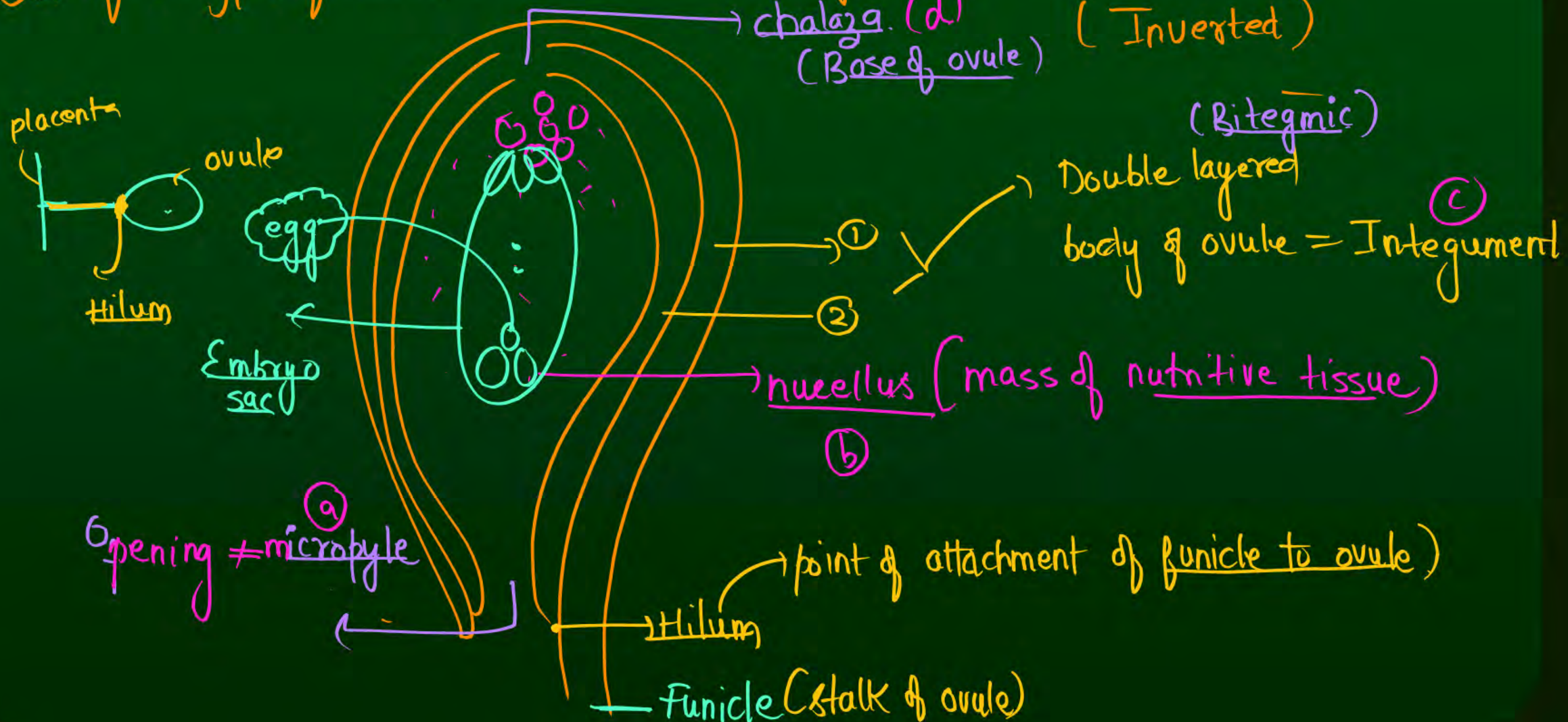
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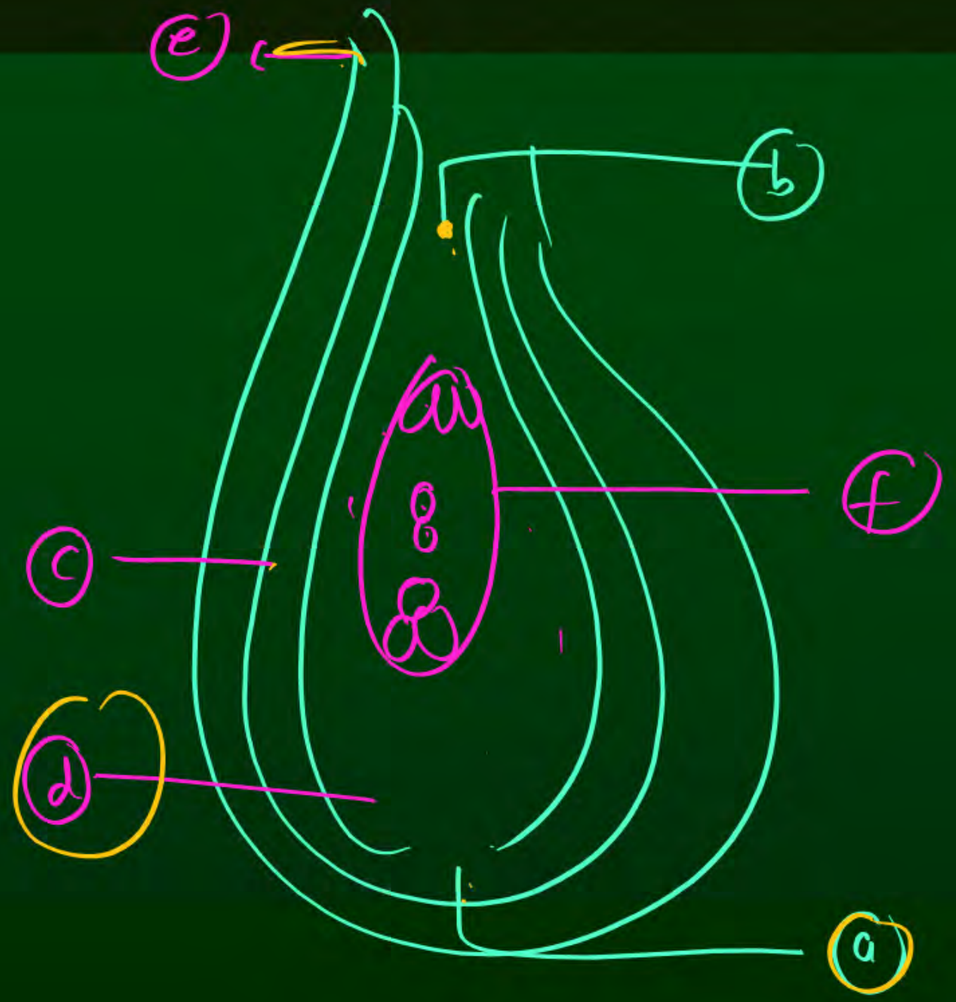
female gamete

(Note:-
Ovary can have
- one ovule → paddy, Mango
- many ovule → papaya, water melon

Structure of ovule

Out of 6 types of ovule, the most common type of ovule is → Anatropous (Inverted)

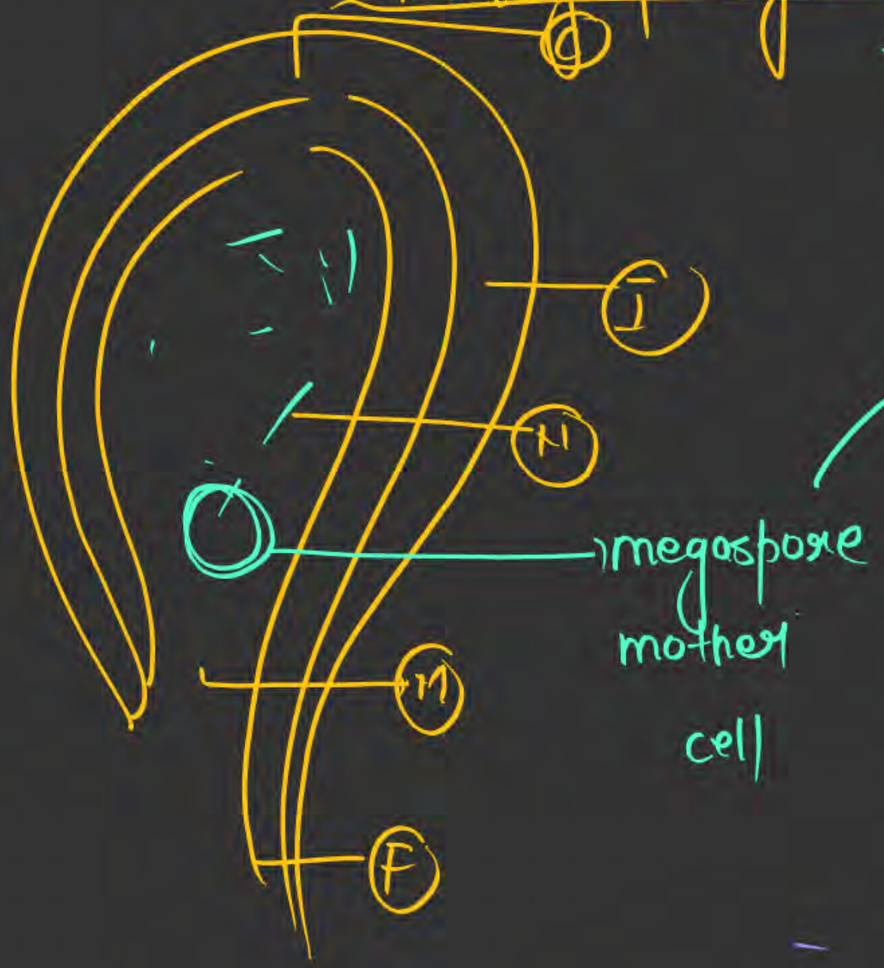




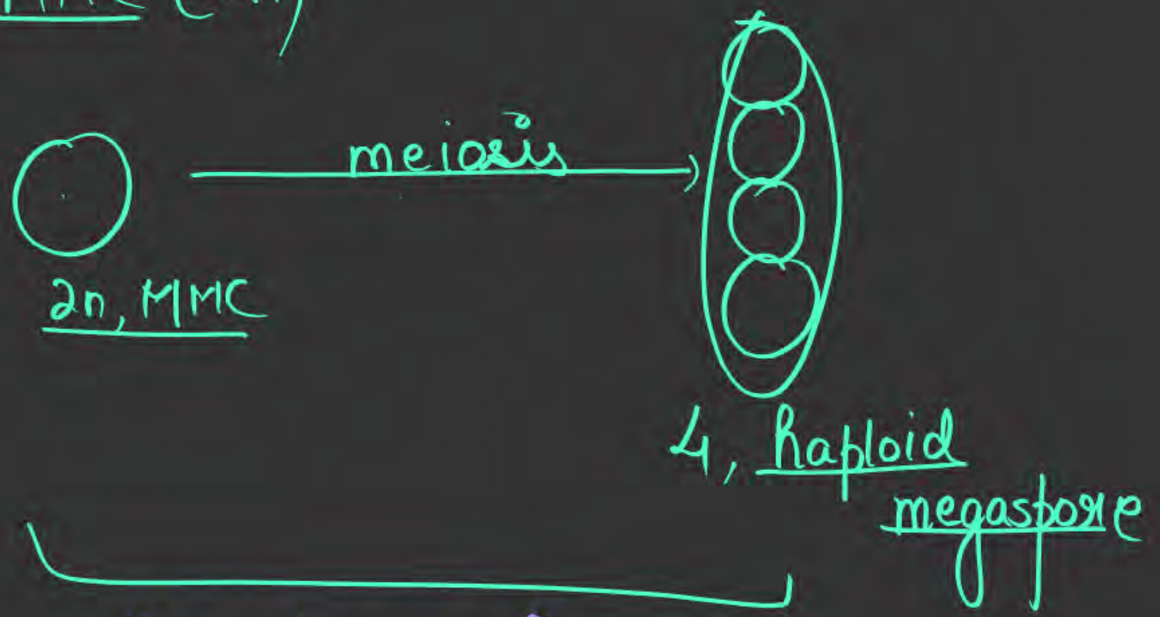
Anatropous

a - chalaza — 2n
b - micropyle — 2n
c - Integument — 2n
d - Nucellus — 2n
e - Funicle — 2n
f - Embryosac — n

Megasporogenesis



- One of the nucellar cell towards micropylar end grows in size and behave as MMC (2n)



(Megasporogenesis)

Megasporogenesis

1.2.2 The Pistil, Megasporangium (ovule) and Embryo sac

The gynoecium represents the female reproductive part of the flower. The gynoecium may consist of a single pistil (**monocarpellary**) or may have more than one pistil (**multicarpellary**). When there are more than one, the pistils may be fused together (**syncarpous**) (Figure 1.7b) or may be free (**apocarpous**) (Figure 1.7c). Each pistil has three parts (Figure 1.7a), **the stigma, style and ovary**. The **stigma** serves as a landing platform for pollen grains. The style is the elongated slender part beneath the stigma. The basal bulged part of the pistil is the **ovary.** Inside the ovary is the **ovarian cavity (locule)**. **The placenta** is located inside the ovarian cavity. Recall the definition and types of placentation that you studied in

of the nucellus. It is a large cell containing dense cytoplasm and a prominent nucleus. The MMC undergoes meiotic division. *What is the importance of the MMC undergoing meiosis?* Meiosis results in the production of four **megaspores** (Figure 1.8a).

Female gametophyte : In a majority of flowering plants, one of the megaspores is **functional** while the other three degenerate. Only the **functional megaspore** develops into the **female gametophyte (embryo sac).** This method of embryo sac formation from a single megaspore is termed **monosporic** development. *What will be the ploidy of the cells of the nucellus, MMC, the functional megaspore and female gametophyte?*

Class XI. Arising from the placenta are the **megasporangia**, commonly called **ovules.** The number of ovules in an ovary may be one (wheat, paddy, mango) to many (papaya, water melon, orchids).

The Megasporangium (Ovule) : Let us familiarise ourselves with the structure of a typical angiosperm ovule (Figure 1.7d). The ovule is a small structure attached to the placenta by means of a stalk called **funicle**. The body of the ovule fuses with funicle in the region called **hilum**. Thus, hilum represents the junction between ovule and funicle. Each ovule has one or two protective envelopes called **integuments.** Integuments encircle the nucellus except at the tip where a small opening called the **micropyle** is organised. Opposite the micropylar end, is the **chalaza**, representing the basal part of the ovule.

Enclosed within the integuments is a mass of cells called the **nucellus**. Cells of the nucellus have abundant reserve food materials. Located in the nucellus is the **embryo sac** or **female gametophyte**. An ovule generally has a single embryo sac formed from a megaspore.

Megasporogenesis : The process of formation of megaspores from the **megaspore mother cell** is called **megasporogenesis**. Ovules generally differentiate a single megaspore mother cell (MMC) in the micropylar region

Let us study about the formation of the embryo sac in detail. (Figure 1.8b). The nucleus of the functional megaspore divides mitotically to form two nuclei which move to the opposite poles, forming the **2-nucleate** embryo sac. Two more sequential mitotic nuclear divisions result in the formation of the **4-nucleate** and later the **8-nucleate** stages of the embryo sac. It is of interest to note that these mitotic divisions are strictly free nuclear, that is, nuclear divisions are not followed immediately by cell wall formation. After the 8-nucleate stage, cell walls are laid down leading to the organisation of the typical **female gametophyte** or **embryo sac**. Observe the distribution of cells inside the embryo sac (Figure 1.8b, c). Six of the eight nuclei are surrounded by cell walls and organised into cells; the remaining two nuclei, called polar nuclei are situated below the egg apparatus in the large **central cell.**

There is a characteristic distribution of the cells within the embryo sac. Three cells are grouped together at the micropylar end and constitute the **egg apparatus**. The egg apparatus, in turn, consists of two **synergids** and one **egg cell**. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the **antipodals**. The large central cell, as mentioned earlier, has two polar nuclei. Thus, a typical angiosperm embryo sac, at maturity, though **8-nucleate** is **7-celled**.

QUESTION

A typical angiospermic anther is bilobed and tetragonal consisting of

1. Two microsporangia
2. Three microsporangia
3. Four microsporangia
4. Only one microsporangium





QUESTION

Select the odd one out w.r.t. wall layers of microsporangium in flowering plants

1. Endothecium
2. Middle layers
3. Tapetum
4. Integument ✓ Epidermis



QUESTION

Pollen grains are well-preserved as fossils because of the presence of

(NEET- 2016)

1. Sporopollenin (exine)
2. Cellulose
3. Pectin
4. Carotenoids



QUESTION

Which of the following option about tapetum is correct?

1. Nutritive tissue
2. Sporogenous tissue
3. Protective and haploid tissue
4. External layer of microsporangium wall



QUESTION

The prominent pollen grain apertures called germ pores are present on

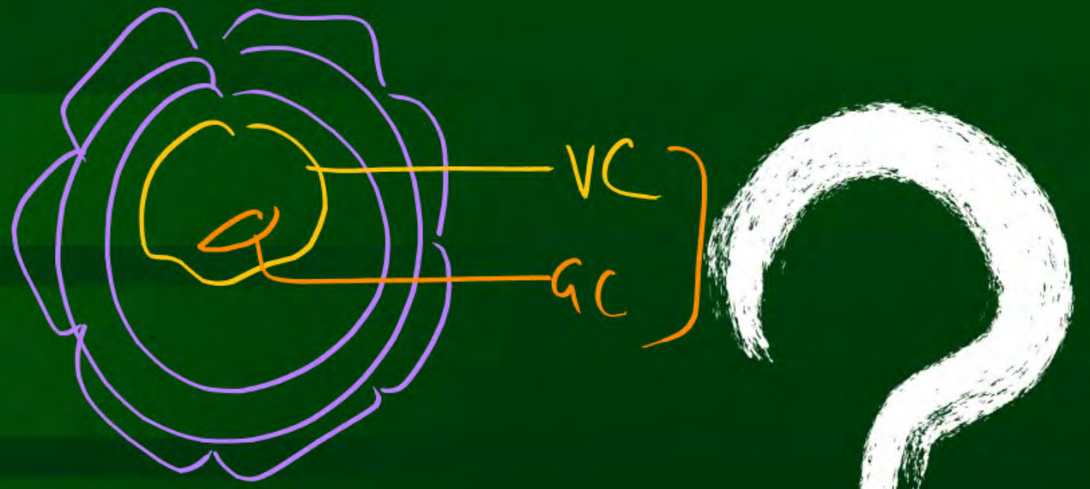
1. Vegetative cell
2. Intine
3. Exine
4. Generative cell



QUESTION

Two-celled stage of pollen grains is the result of

1. Meiosis
2. Symmetric mitosis
3. Asymmetric mitosis
4. Amitosis



QUESTION

In most of the flowering plants, pollination takes place at _____ celled stage.

60% - 2-celled
40% - 3-cell.

1. 2
2. 3
3. 4
4. 5



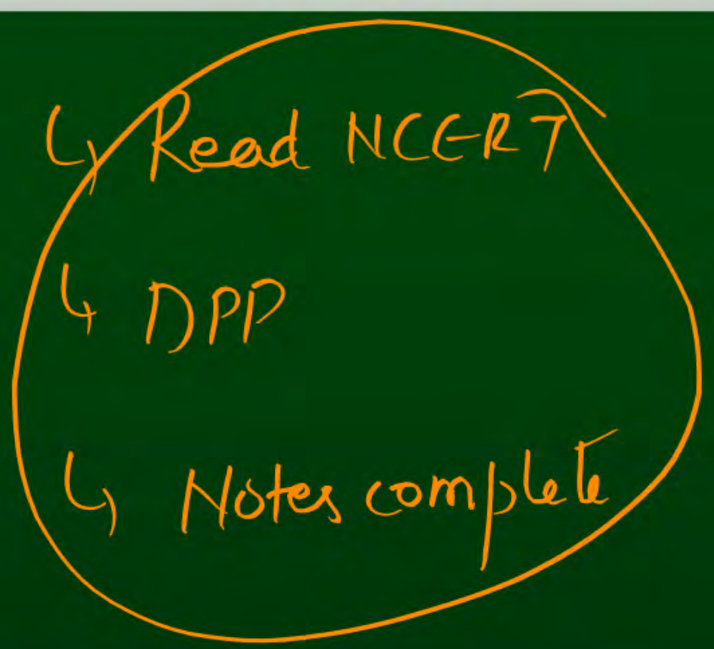
QUESTION

What is incorrect for generative cell?

1. Floats in cytoplasm of vegetative cell
2. Spindle-shaped
3. Having abundant food reserve
4. as dense cytoplasm and a nucleus



Homework from Parishram 2.0 Module



THANK YOU